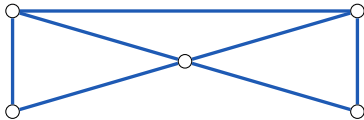


Introductory Concepts

Section 1.1

Harris–Hirst–Mossinghoff, *Combinatorics and Graph Theory*



Jeremy Quastel

Today's lecture

1. History

The Königsberg Bridge Problem
and the origins of graph theory.

3. The Basics

Section 1.1.2.

2. Graphs and Their Relatives

Section 1.1.1.

4. Special Types of Graphs

Section 1.1.3.

The Königsberg Bridge Problem

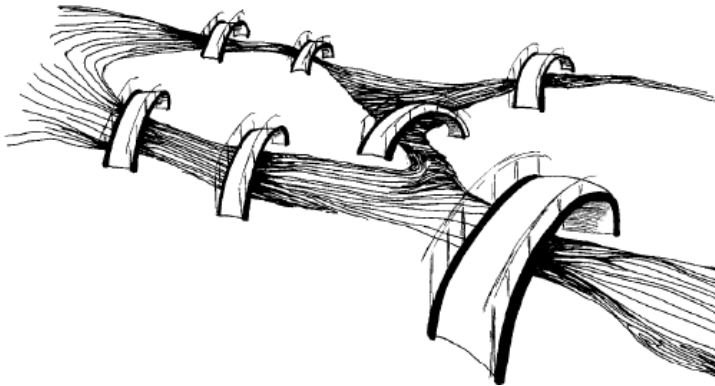


Figure 1.1: The bridges in Königsberg.

The Pregolya River passed through the city of Königsberg, where seven bridges connected four regions of land. The city's residents enjoyed walking across the bridges, but no one could find a route that crossed each bridge exactly once.

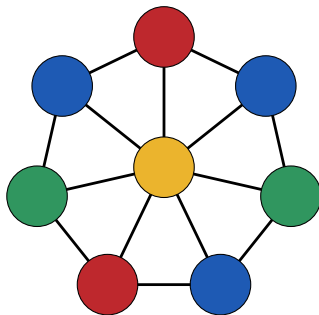
Source: Harris, Hirst, and Mossinghoff, *Combinatorics and Graph Theory*, 2nd ed., Figure 1.1.

The Four-Colour Map Conjecture

Conjecture

Four is the maximum number of colors required to color any map where bordering regions are colored differently.

Example: The following is a four-colouring of a graph with 8 vertices.



Introductory Concepts

Basic Definitions

Graph, vertices, and edges

A **graph** consists of two finite sets, V and E .

The elements of V are called **vertices**.

The elements of E are called **edges**. Each edge is an unordered pair of vertices.

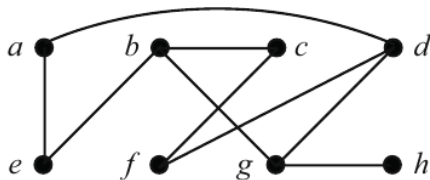


FIGURE 1.2. A visual representation of the graph G .

Example

$$V = \{a, b, c, d, e, f, g, h\},$$

$$E = \{\{a, d\}, \{a, e\}, \{b, c\}, \{b, e\}, \{b, f\}, \{b, g\}, \{c, f\}, \{c, g\}, \{d, f\}, \{d, g\}, \{g, h\}\}.$$

Together, V and E form the graph G shown in Figure 1.2.

Graphs and Their Relatives

We can obtain similar structures by altering our definition in various ways.

Example: A directed graph

By replacing our set E with a set of ordered pairs of vertices, we obtain a **directed graph**, or **digraph**. Each edge of a digraph has a specific orientation.

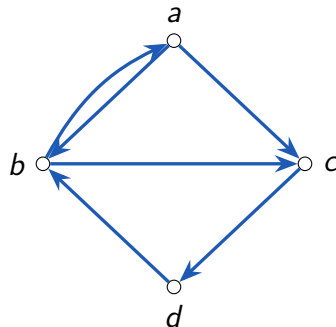


Figure 1.3: A digraph.

Source: Harris, Hirst, and Mossinghoff, *Combinatorics and Graph Theory*, 2nd ed., Figure 1.3.

Next Example:

A multigraph

If we allow repeated elements in our set of edges, technically replacing our set E with a multiset, we obtain a **multigraph**.

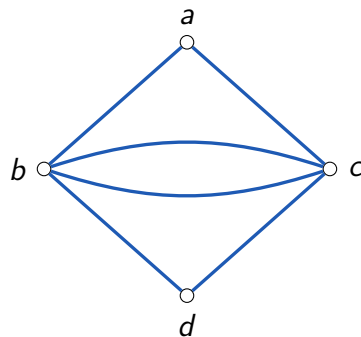


Figure 1.4: A multigraph.

Source: Harris, Hirst, and Mossinghoff, *Combinatorics and Graph Theory*, 2nd ed., Figure 1.4.

Next Example:

A pseudograph

By allowing edges to connect a vertex to itself (“loops”), we obtain a **pseudograph**.

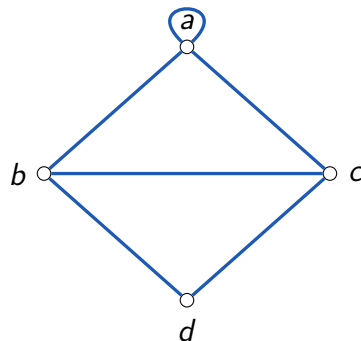


Figure 1.5: A pseudograph.

Source: Harris, Hirst, and Mossinghoff, *Combinatorics and Graph Theory*, 2nd ed., Figure 1.5.

Next Example:

A hypergraph

Allowing our edges to be arbitrary subsets of vertices, rather than just pairs, gives us **hypergraphs**.

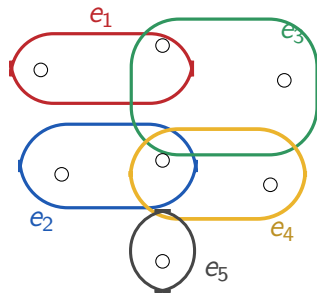


Figure 1.6: A hypergraph with 7 vertices and 5 edges.

Source: Harris, Hirst, and Mossinghoff, *Combinatorics and Graph Theory*, 2nd ed., Figure 1.6.

Next Example:

An infinite graph

By allowing V or E to be an infinite set, we obtain **infinite graphs**. Infinite graphs are studied in Chapter 3.



An infinite path.

Back to Graphs

Vertex set

The vertex set of G , denoted by $V(G)$, is the set of its vertices.

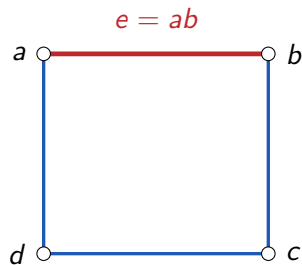
$$V(G) = \{a, b, c, d\}.$$

Edge set

The edge set of G , denoted by $E(G)$, is the set of its edges.

$$E(G) = \{ab, bc, cd, ad\}.$$

Here ab is shorthand for the unordered pair $\{a, b\}$.



Graph G

Definition: Adjacent and Incident Vertices

Adjacent: Vertices u, v are adjacent if $uv \in E(G)$.

Example: a and b are adjacent.

Nonadjacent: Vertices u, v are nonadjacent if $uv \notin E(G)$.

Example: a and c are nonadjacent.

End vertices: The end vertices of an edge uv are u and v .

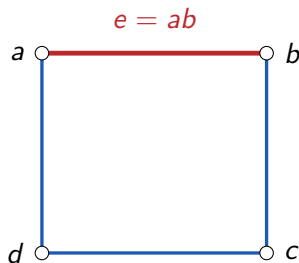
Example: a and b are the end vertices of $e = ab$.

Incident: A vertex is incident with an edge if it is one of its end vertices.

Example: a is incident with e , but c is not.

Order: The order of G is $|V(G)|$, the number of vertices.

Example: G has order 4.



Graph G

Definition: Neighborhood

Neighborhood (open neighborhood):

$N(v) = \{x \in V(G) : vx \in E(G)\}$ is the set of vertices adjacent to v .

Example: $N(c) = \{a, b, d\}$.

Closed neighborhood:

$N[v] = \{v\} \cup N(v)$.

Example: $N[c] = \{a, b, c, d\}$.

Neighborhood of a set:

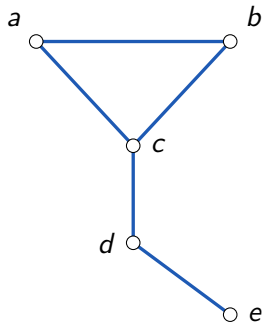
For $S \subseteq V(G)$, $N(S) = \bigcup_{v \in S} N(v)$.

Example: For $S = \{a, d\}$, $N(S) = \{b, c, e\}$.

Closed neighborhood of a set:

$N[S] = S \cup N(S)$.

Example: $N[\{a, d\}] = \{a, b, c, d, e\}$.



Graph G

Definition: Degree

Degree of a vertex:

$\deg(v)$ is the number of edges incident with v .

For a simple graph, $\deg(v) = |N(v)|$.

Example: $\deg(c) = 3$ and $\deg(e) = 1$.

Maximum degree:

$\Delta(G) = \max\{\deg(v) : v \in V(G)\}$.

Example: $\Delta(G) = 3$, attained at c .

Minimum degree:

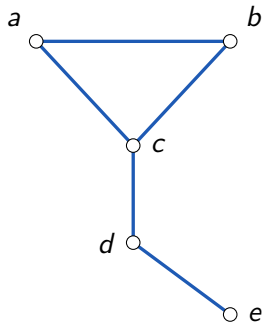
$\delta(G) = \min\{\deg(v) : v \in V(G)\}$.

Example: $\delta(G) = 1$, attained at e .

Degree sequence:

The list of all vertex degrees, usually in descending order.

Example: $(3, 2, 2, 2, 1)$.



Graph G

First Theorem of Graph Theory

Theorem

In a graph G , the sum of the vertex degrees equals twice the number of edges:

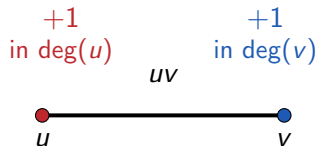
$$\sum_{v \in V(G)} \deg(v) = 2|E(G)|.$$

Consequently, the number of vertices of odd degree is even.

Proof

When we sum the degrees, each edge uv is counted once at u and once at v . Thus every edge contributes 2, giving the identity.

The sum is even, as is the contribution from even degrees. Hence the number of odd degrees is even. $\square$



$$1 + 1 = 2$$

Each edge is counted twice.

Definitions: Walks and Paths

Walk:

A sequence $v_1, \dots, v_k$ with $v_i v_{i+1} \in E(G)$ for $1 \leq i < k$.

Vertices and edges may repeat.

Example: a, b, a, c, d, e .

Path:

A walk in which all vertices are distinct.

Example: a, b, c, d, e .

End vertices:

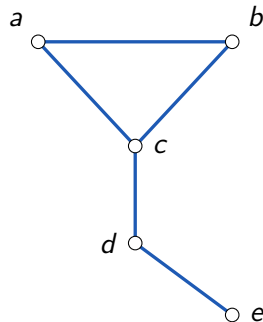
The first and last vertices of a walk $v_1, \dots, v_k$ are its end vertices. It is a v_1-v_k walk.

Example: a, b, c, d, e has end vertices a and e .

Trail:

A walk in which all edges are distinct.

Example: c, a, b, c, d, e repeats c , but no edge.



Graph G

Definitions: Close Paths

Closed path / Cycle:

A path $v_1, \dots, v_k$, with $k \geq 3$, together with the edge $v_k v_1$ joining its last vertex to its first.

Example: a, b, c, a .

Closed trail / Circuit:

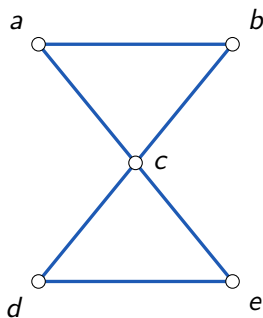
A trail that begins and ends at the same vertex.

Example: c, a, b, c, d, e, c repeats c , but no edge.

Length of a walk:

The number of edges traversed, counting repetitions. The same definition applies to paths, trails, and cycles.

Example: a, b, a, c, d has length 4; the edge ab is counted twice.



Graph G

Second Theorem of Graph Theory

Theorem 1.2

In a graph G with vertices u and v , every $u-v$ walk contains a $u-v$ path.

Proof

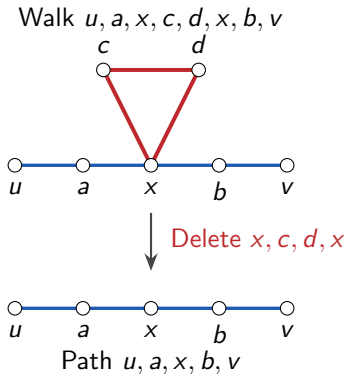
We induct on the length k of $W = (u = w_0, w_1, \dots, w_k = v)$. The case $k = 0$ is immediate.

If all vertices are distinct, W is already a path.

Otherwise, choose $j < r$ with $w_j = w_r$. Delete the closed portion from w_j to w_r , leaving

$$W_1 : w_0, \dots, w_j, w_{r+1}, \dots, w_k.$$

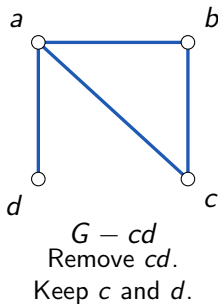
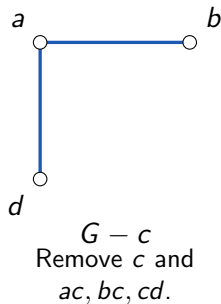
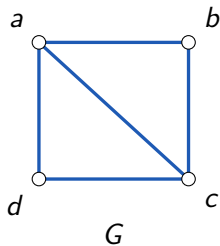
This shorter $u-v$ walk contains a $u-v$ path by induction. That path also lies in W . $\square$



Vertex and Edge Deletion

Vertex deletion: $G - v$ is the graph obtained by removing v and every edge incident with v .

Edge deletion: $G - e$ is the graph obtained by removing only the edge e , keeping its end vertices.

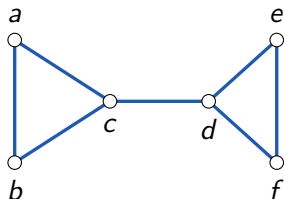


Definition: Connected Graphs

Connected: A graph is connected if every pair of vertices can be joined by a path.

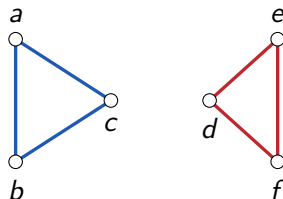
Disconnected: A graph is disconnected if some pair of vertices cannot be joined by a path.

Connected component: A maximal connected subgraph, containing all vertices and edges that can be reached from any one of its vertices.



Connected

One connected component.



Disconnected

Two components: $\{a, b, c\}$ and $\{d, e, f\}$.

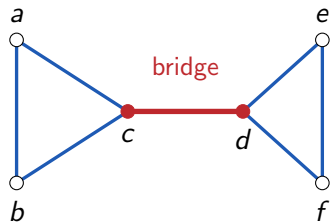
Definition: Cuts and Bridges

Cut vertex: A vertex v is a cut vertex if $G - v$ has more connected components than G .

Example: Both c and d are cut vertices. Deleting either one splits G into two components.

Bridge: An edge e is a bridge if $G - e$ has more connected components than G .

Example: The edge cd is a bridge. Deleting it separates $\{a, b, c\}$ from $\{d, e, f\}$.

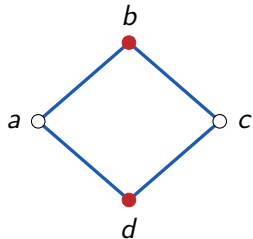


Cut vertices: c, d

Graph G

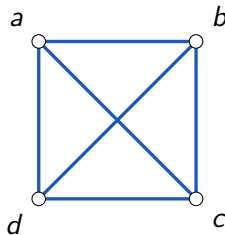
Definition: Complete Graphs

Vertex cut set: A proper subset $S \subsetneq V(G)$ is a vertex cut set if $G - S$ is disconnected.



$S = \{b, d\}$ is a vertex cut set.
Deleting S leaves isolated vertices a, c .

Complete graph: A graph is complete if every pair of distinct vertices is adjacent. We write K_n for the complete graph on n vertices.



K_4 : all six possible edges are present.

Question: Do complete graphs have vertex cut sets?

Question

Question

Let G be a complete graph. Can deleting a proper subset $S \subsetneq V(G)$ disconnect G ? In other words, does G have a vertex cut set?

Question

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Let G be a complete graph. Can deleting a proper subset $S \subsetneq V(G)$ disconnect G ? In other words, does G have a vertex cut set?

Answer

No, since $G - S$ is connected for every proper subset $S \subsetneq V(G)$.

Definition: Connectivity

Connectivity: For a non-complete graph G , the connectivity $\kappa(G)$ is the minimum size of a vertex cut set.

If G is disconnected, $\kappa(G) = 0$. If G is connected and non-complete, with n vertices, then

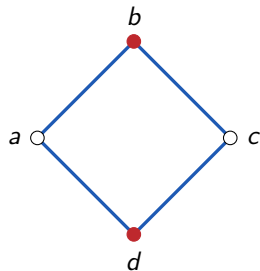
$$1 \leq \kappa(G) \leq n - 2.$$

Complete graphs: By convention, $\kappa(K_n) = n - 1$. Thus, for every graph of order $n \geq 1$,

$$0 \leq \kappa(G) \leq n - 1,$$

and $\kappa(G) = n - 1$ if and only if G is complete.

k -connected: For a positive integer k , a graph G is k -connected if $\kappa(G) \geq k$.



Example: $\kappa(G) = 2$

Deleting any one vertex leaves a connected graph.

Deleting $\{b, d\}$ disconnects it.

So G is 2-connected.

Prove the Following

Statement

For a graph G with at least two vertices, G is connected if and only if $\kappa(G) \geq 1$.

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For a graph G with at least two vertices, G is connected if and only if $\kappa(G) \geq 1$.

If: $\kappa(G) \geq 1 \implies G$ is connected

If G were disconnected, no vertex deletion would be needed to disconnect it, so $\kappa(G) = 0$.
Thus $\kappa(G) \geq 1$ implies that G is connected.

Prove the Following

Statement

For a graph G with at least two vertices, G is connected if and only if $\kappa(G) \geq 1$.

If: $\kappa(G) \geq 1 \implies G$ is connected

If G were disconnected, no vertex deletion would be needed to disconnect it, so $\kappa(G) = 0$. Thus $\kappa(G) \geq 1$ implies that G is connected.

Only if: G is connected $\implies \kappa(G) \geq 1$

If G is connected and non-complete, deleting no vertices leaves it connected. Every vertex cut set therefore has at least one vertex, so $\kappa(G) \geq 1$.

If $G = K_n$, then $\kappa(G) = n - 1 \geq 1$ because $n \geq 2$.

Prove the Following

Statement

For a graph G with at least three vertices, $\kappa(G) \geq 2$ if and only if G is connected and has no cut vertices.

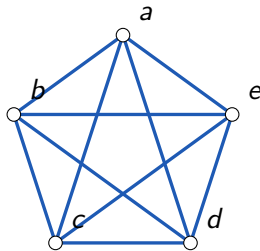
Other Things You Can Prove

- (ii) For a graph G with at least three vertices, $\kappa(G) \geq 2$ if and only if G is connected and has no cut vertices.
- (iii) Every 2-connected graph contains at least one cycle.
- (iv) For every graph G , $\kappa(G) \leq \delta(G)$.

Special Types of Graphs

Complete graphs

The complete graph K_n has n vertices, with an edge between every pair of distinct vertices.

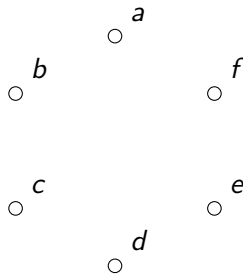


Example of a complete graph: K_5

Special Types of Graphs

Empty graphs

The empty graph E_n has n vertices and no edges. Its edge set is $\emptyset$.

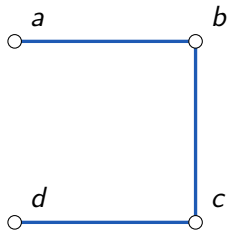


Example of an empty graph: E_6

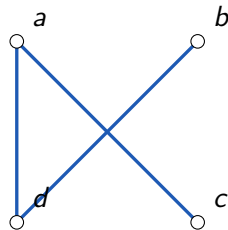
Special Types of Graphs

Complements

The complement $\overline{G}$ has the same vertices as G . Two distinct vertices are adjacent in $\overline{G}$ exactly when they are not adjacent in G .



The original graph G

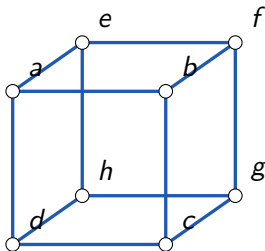


The complement $\overline{G}$

Special Types of Graphs

Regular graphs

A graph is regular if every vertex has the same degree. It is r -regular if $\deg(v) = r$ for every vertex v .

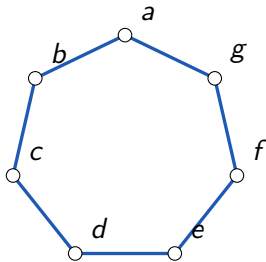


Example of a regular graph: every vertex has degree 3

Special Types of Graphs

Cycles

The cycle graph C_n , for $n \geq 3$, consists of a cycle on n vertices. It has n edges.

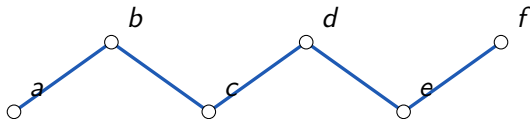


Example of a cycle graph: C_7

Special Types of Graphs

Paths

The path graph P_n consists of a path on n vertices. It has $n - 1$ edges.

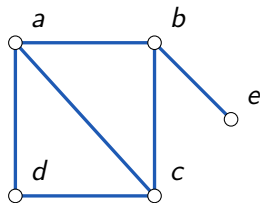


Example of a path graph: P_6

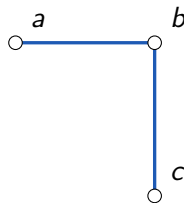
Special Types of Graphs

Subgraphs

A graph H is a subgraph of G if $V(H) \subseteq V(G)$ and $E(H) \subseteq E(G)$. We write $H \subseteq G$.



The original graph G

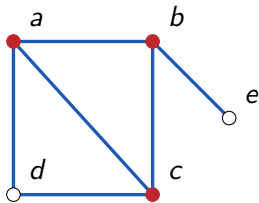


Example of a subgraph H of G

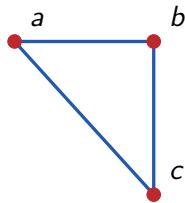
Special Types of Graphs

Induced subgraphs

For $S \subseteq V(G)$, the induced subgraph $G[S]$ has vertex set S and includes every edge of G with both end vertices in S .



Graph G with selected vertices $S = \{a, b, c\}$

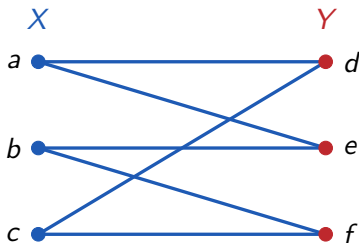


The induced subgraph $G[S]$

Special Types of Graphs: Bipartite Graphs

Bipartite graphs

A graph is bipartite if its vertex set can be partitioned into two disjoint sets X and Y such that every edge has one end vertex in X and the other in Y . The sets X and Y are called the partite sets.



Example of a bipartite graph with partite sets
 $X = \{a, b, c\}$ and $Y = \{d, e, f\}$.

Theorem: Bipartite Graphs and Odd Cycles

Theorem 1.3

A graph with at least two vertices is bipartite if and only if it contains no odd cycles.

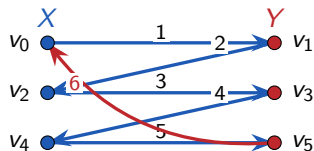
Proof of Theorem

Theorem 1.3

A graph with at least two vertices is bipartite if and only if it contains no odd cycles.

Proof: Forward direction

Let G be bipartite with partite sets X and Y . Along a cycle, successive vertices alternate between X and Y . Starting in X , we are in X after an even number of edges and in Y after an odd number. Returning to the starting vertex therefore requires an even number of edges.



Each step switches sides. Hence, you will need to do an even number of steps to return to the same side. In particular, you will need to do an even number of steps to return to the same vertex.

Proof of Theorem

Theorem 1.3

A graph with at least two vertices is bipartite if and only if it contains no odd cycles.

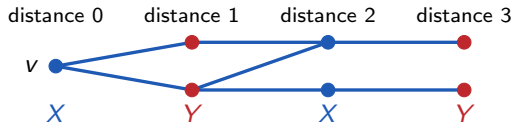
Proof: Reverse direction

Suppose G has no odd cycles. We may assume G is connected, treating each connected component separately otherwise. Fix a vertex v and define

$$X = \{u \in V(G) : d(v, u) \text{ is even}\}, \quad Y = \{u \in V(G) : d(v, u) \text{ is odd}\}.$$

Here $d(v, u)$ is the length of a shortest path from v to u .

It suffices to show that no two vertices in X are adjacent, and similarly for Y . Since X and Y partition $V(G)$, every edge must then have exactly one end vertex in each set. Thus G is bipartite.



Proof of Theorem

Theorem 1.3

A graph with at least two vertices is bipartite if and only if it contains no odd cycles.

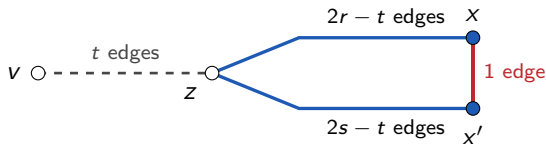
Proof: Reverse direction continued

We argue by contradiction. Suppose $x, x' \in X$ are adjacent. Choose shortest paths from v to x and from v to x' , of lengths $2r$ and $2s$. Let z be their last common vertex and put $t = d(v, z)$.

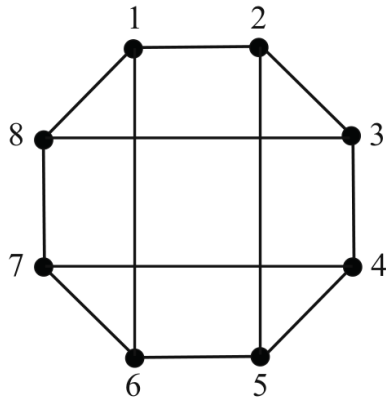
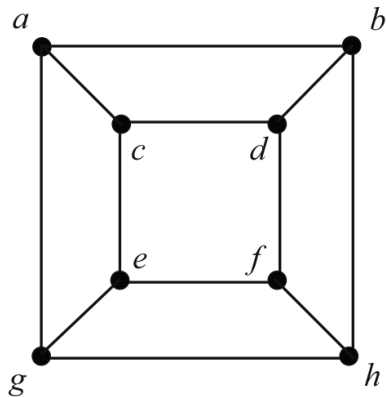
The portions from z to x and x' meet only at z . Together with xx' , they form a cycle of length

$$(2r - t) + (2s - t) + 1 = 2(r + s - t) + 1,$$

which is odd, a contradiction. The same argument applies to two vertices in Y .



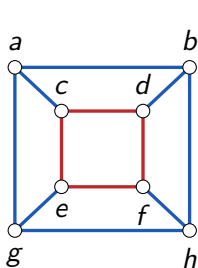
Graph Isomorphisms



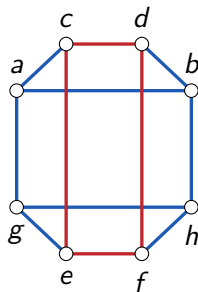
Are these graphs the same?

Source: Harris, Hirst, and Mossinghoff, *Combinatorics and Graph Theory*, 2nd ed., Figure 1.21.

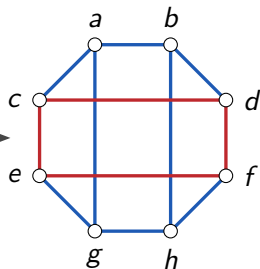
Are the Graphs the Same: Yes



Original drawing



Make the inner rectangle
taller than the outer one



Widen the tall rectangle
and narrow the other one

Every edge keeps the same end vertices throughout.

To match the book's labels: $a \mapsto 1, b \mapsto 2, c \mapsto 8, d \mapsto 3,$
 $e \mapsto 7, f \mapsto 4, g \mapsto 6, h \mapsto 5.$

Definition: Graph Isomorphism

Graph isomorphism

Graphs G and H are **isomorphic** if there exists a one-to-one correspondence

$$f : V(G) \longrightarrow V(H)$$

such that, for every pair of vertices $x, y \in V(G)$,

$$xy \in E(G) \iff f(x)f(y) \in E(H).$$

The mapping f is called an **isomorphism**.

Example: For the two graphs on the previous slides, an isomorphism is

x	a	b	c	d	e	f	g	h
$f(x)$	1	2	8	3	7	4	6	5

For example, the edge ac corresponds to the edge 18.

Graph Isomorphisms Continued

Properties preserved by an isomorphism

If G and H are isomorphic, then they have:

- The same number of vertices and the same number of edges.
- The same degree at corresponding vertices, hence identical degree sequences.
- The same connectivity and the same numbers of cycles of each length.

Their complements $\overline{G}$ and $\overline{H}$ are also isomorphic, using the same vertex correspondence.

Matching vertex counts, edge counts, and degree sequences alone does not guarantee that two graphs are isomorphic.

In general, determining whether two graphs are isomorphic can be difficult, especially as the graphs become large.

Next time: Distance in Graphs